Reconstructing Phosphorus Dynamics in the Precambrian Ocean Using Independent Component Analysis of Shale Geochemistry

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Background: Phosphorus in the Precambrian

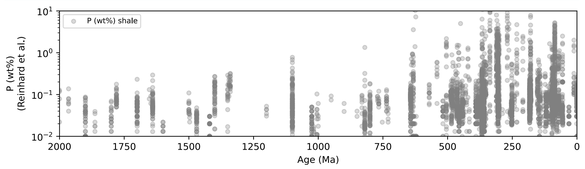


Figure 1. P(wt%) in shales. P content rises substantially from ca. 650Ma[1].

- **P** is an ultimate limiting nutrient[2]. Low Precambrian pO_2 → $[P]_{seawater}$ scarcity in the Precambrian[3]?
- Compilations of shale P concentrations appear to support low [P] before ~650Ma.

However

- Shale is a mixture of several phases: detritus, authigenic apatite, organic, **Fe-oxide-bound** etc. →Changes in bulk shale P do not directly reflect that of $[P]_{seawater}$.
- Fe-oxide-bound P could record $[P]_{seawater}$[4].

Research Q: Secular variation in Fe-bound P

If we isolate only the Fe-oxide-bound P (P/Fe) from the shale dataset, what does the Precambrian record look like?

Was the Precambrian ocean truly depleted in [P]?

Approach : Why ICA?

PCA (Principal Component Analysis)
- Find axes that maximize the variance
- Vectors must be **orthogonal**

ICA (Independent Component Analysis)
- Find "statistically independent" axes by maximizing non-Gaussianity

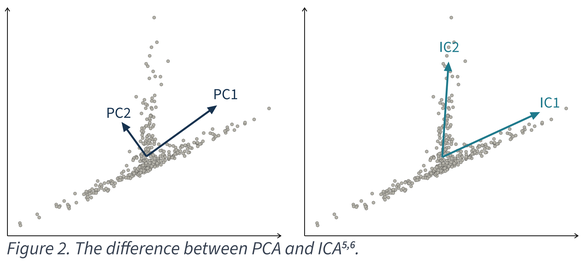


Figure 2. The difference between PCA and ICA[5,6].

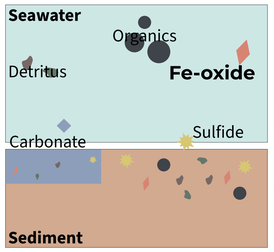


Figure 3. Schematics of P carriers from seawater to sediment.

Fe-oxide P tracks seawater P

$$K_{adsorption} = \frac{P/Fe(III)}{[P]_{seawater}}$$

Each sample is a point in element space
x = standardized composition

ICA finds trend vectors
w = direction of max non-Gaussianity

Score = position along the trend
$s = \mathbf{w}^T\mathbf{x}$
A high score on a vector means the sample is rich in that component

Dataset

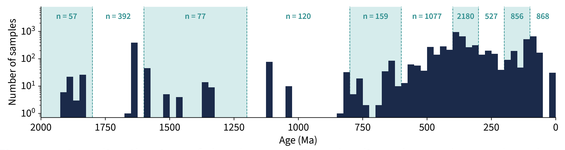


Figure 4. Age distribution of the dataset. The shading shows the analyzed age bins

- Bulk shale geochemistry from the Sedimentary Geochemistry and Paleoenvironments Project (**SGP**)[7].
- 13 variables representing the main sedimentary phases: TOC, Mg, P, S, Ca, V, Cr, Mn, Fe, Zn, Mo, U, Zr
- The dataset is divided into 10 age bins, with ICA applied to each bin separately.

ICA Workflow

Step 1 · Pre-processing
- Z-score standardization of each element

Step 2 · Run ICA
- **FastICA** is non-deterministic: results change with the random seed
- ICA run 100 times, for every number of components

Step 3 · Hierarchical clustering → choose n
- IC vectors from 100 runs clustered by absolute cosine distance (sign-independent)
- n chosen as the largest number below the instability peak that still gives tight, and interpretable clusters
- Stable ICs = cluster centers at the chosen n

Step 4 · Assign ICs to phases
- Each IC identified from its element loadings, e.g. Fe + Mn → Fe-oxide, Fe + S → sulfide

Step 5 · Extract the P signal
- An IC is a direction of enrichment, not a pure end-member
- Samples in the top 25% of Fe-oxide IC scores → median P/Fe as a proxy for seawater DIP

Results & Discussion

Example: **600-400Ma window**

- For each n, ICA was run 100 times.
- Components were clustered by 1-|cosθ|, and the y-axis shows the largest linkage distance needed to recover n clusters.
- 0 means every run returns identical components.
- **n=6 was chosen as the robust n**

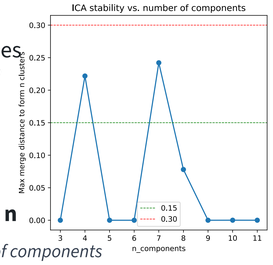


Figure 5. Clustering stability vs. number of components

- Each block = one clustered IC recovered across 100 runs; yellow = identical direction (|cosθ|≈1).
- At n=6, perfectly block-diagonal (not shown)
- At n=7, several clusters are noisy and overlap.

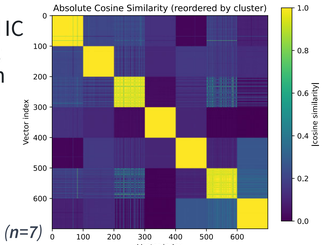
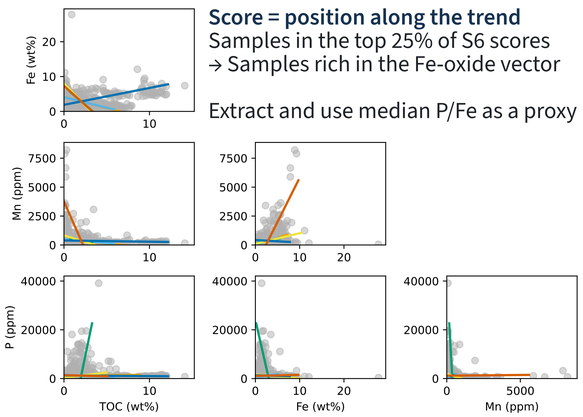


Figure 6. Cosine-similarity heatmap (n=7)

*Figure 7. Element loadings of the six stable ICs. **S6 (Fe+Mn)** is the Fe-oxide component used to extract P/Fe.*



Score = position along the trend
Samples in the top 25% of S6 scores → Samples rich in the Fe-oxide vector

Extract and use median P/Fe as a proxy

Figure 8. Crossplots of TOC, Fe, Mn, and P.

KEY FINDING

Fe-oxide-bound P in shales shows no Precambrian P deficit: Proterozoic P/Fe equals or exceeds the Phanerozoic values.

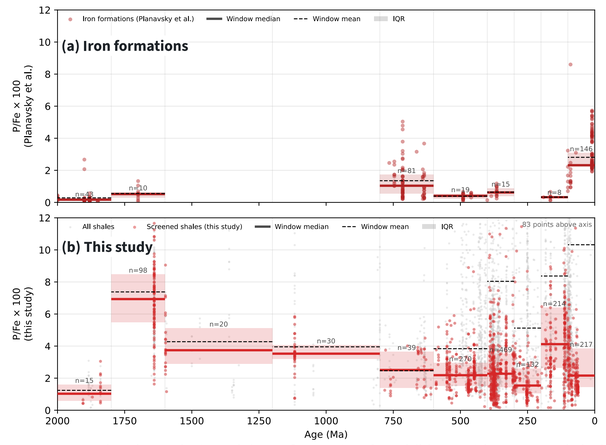


Figure 9. P/Fe through time. (a)Iron formations[8] and (b)Fe-oxide-screened shales (this study; red = top 25% of Fe-oxide IC scores, gray = all shales). Bars show window medians with IQR.

- Shales extend the record into 1600–800 Ma, where iron-formations are absent.
- The apparent Precambrian P scarcity in shale records likely reflects phase mixing, not seawater chemistry.
- Iron formations record 5-20× lower P/Fe: possibly due to Si suppressing P sorption[9].
- Caveat: some bins are dominated by a single locality.

Conclusions

1. ICA isolates a stable Fe-oxide component in shales.
2. Fe-oxide-bound P/Fe shows no Precambrian deficit.
3. Low Precambrian P in bulk shale records ≠ low $[P]_{seawater}$.

References

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